

TE Curve Verification Pipeline Best Model Collage Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report compares representative `TE Curve Verification Pipeline` TE-curve predictions for the current best reference, RCIM Model-Bank Reproduction, Wave 1 directional, and Wave 1 global models. Each model is shown as one four-image collage so local oscillation tracking can be inspected directly.

Scope

- each collage contains four deterministic held-out test curves;
- forward models are shown on forward curves only;
- backward models are shown on backward curves only;
- global Wave 1 models are shown on two forward and two backward curves;
- `Measured TE` uses the same line width as predictions and a dark-gray color for balanced visual comparison.

Metrics Summary

Forward Accepted Non PINN Incumbent Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
<code>accepted_periodic_gru_sequence_fw</code>	<code>accepted_non_pinn_incumbent</code>	Fw	0.001618	0.001931	3.278
<code>accepted_periodic_mlp_harmonic_fw</code>	<code>accepted_non_pinn_incumbent</code>	Fw	0.001694	0.002008	3.439

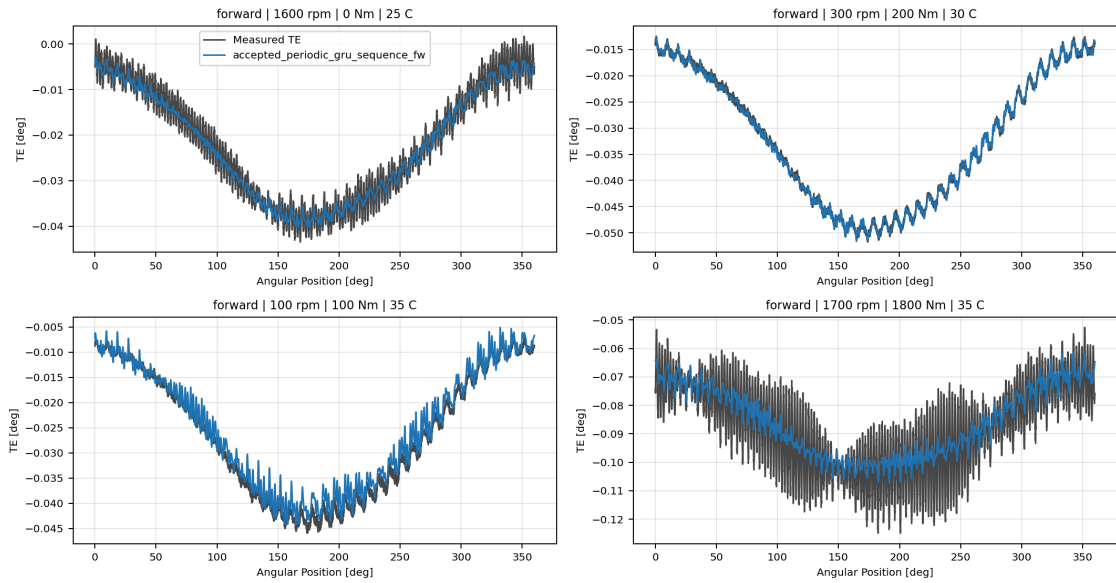
Forward Wave52r Offline Leader Cross Surface Promotion Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
<code>wave52r_promotion_k01_fw_seed_271828</code>	<code>wave52r_offline_leader_cross_surface_promotion</code>	Fw	0.001358	0.001620	2.681
<code>wave52r_promotion_h08_fw_seed_161803</code>	<code>wave52r_offline_leader_cross_surface_promotion</code>	Fw	0.001688	0.001985	3.477

Collage Gallery - Forward Accepted Non PINN Incumbent Models

accepted_periodic_gru_sequence_fw

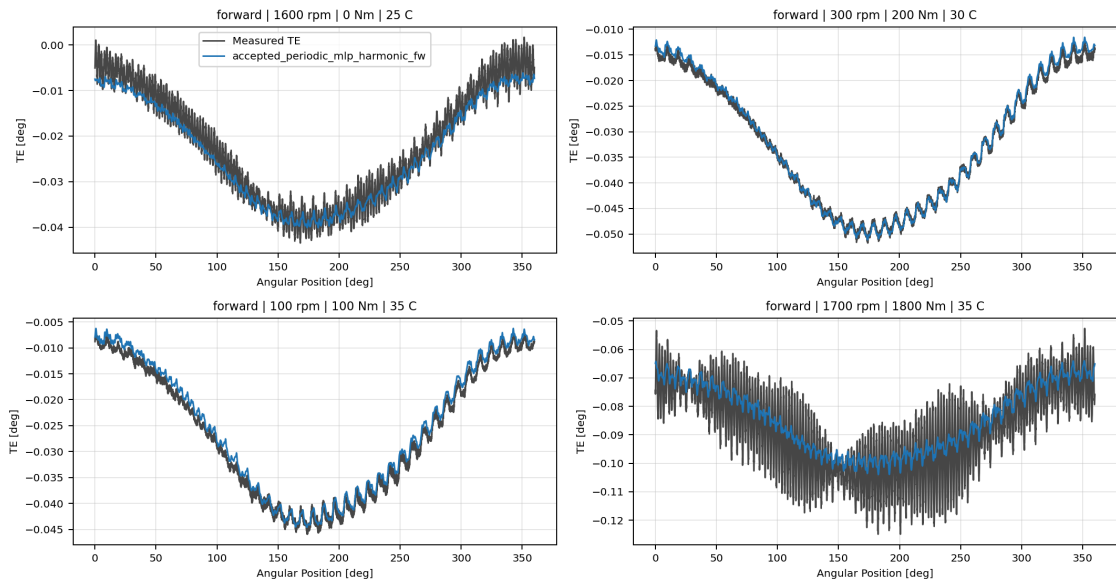
accepted_periodic_gru_sequence_fw



accepted_periodic_gru_sequence_fw TE Curve Verification Pipeline collage

accepted_periodic_mlp_harmonic_fw

accepted_periodic_mlp_harmonic_fw

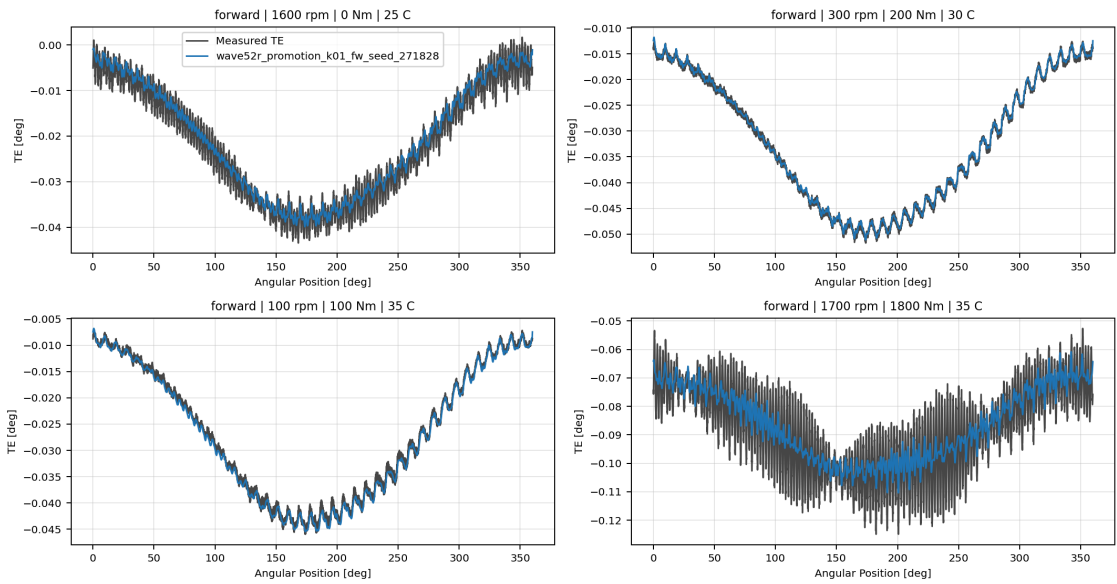


accepted_periodic_mlp_harmonic_fw TE Curve Verification Pipeline collage

Collage Gallery - Forward Wave52r Offline Leader Cross Surface Promotion Models

wave52r_promotion_k01_fw_seed_271828

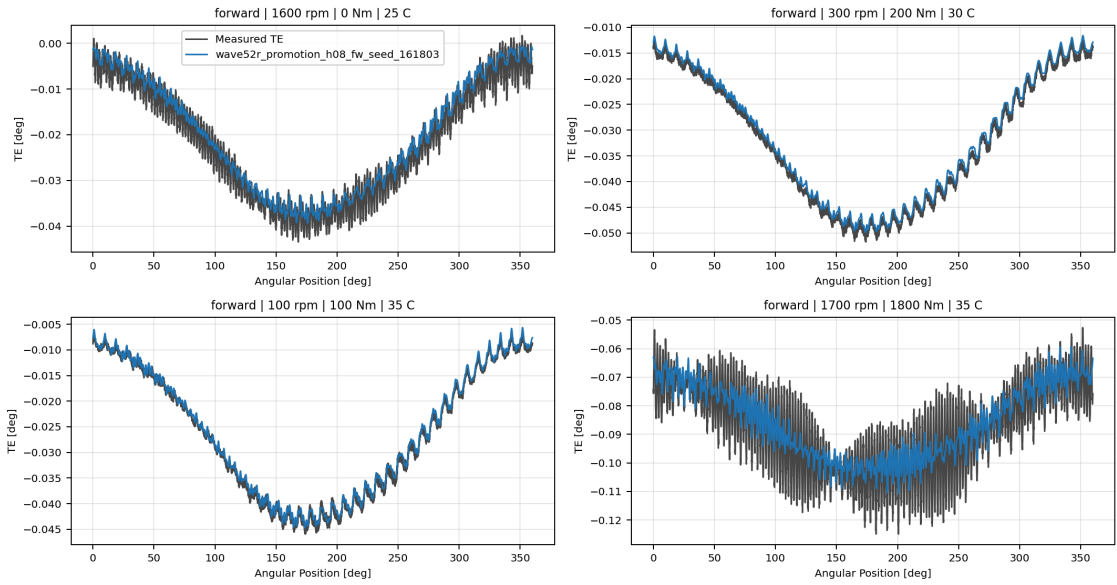
wave52r_promotion_k01_fw_seed_271828



wave52r_promotion_k01_fw_seed_271828 TE Curve Verification Pipeline collage

wave52r_promotion_h08_fw_seed_161803

wave52r_promotion_h08_fw_seed_161803



wave52r_promotion_h08_fw_seed_161803 TE Curve Verification Pipeline collage

Output Artifacts

- output directory:
output\analysis\wave_5_2r\offline_leader_cross_surface_track2\visual_collages\forward\2026-07-31-13-38-10__track2_best_model_collage_report ;
- summary YAML:
output\analysis\wave_5_2r\offline_leader_cross_surface_track2\visual_collages\forward\2026-07-31-13-38-10__track2_best_model_collage_report\track2_best_model_collage_summary.yaml ;
- metrics CSV:
output\analysis\wave_5_2r\offline_leader_cross_surface_track2\visual_collages\forward\2026-07-31-13-38-10__track2_best_model_collage_report\track2_best_model_collage_metrics.csv ;
- report Markdown:
doc\reports\analysis\model_development_waves\wave_5_2\offline_leader_global_promotion\visual_collages\forward\[2026-07-31]\track2_best_model_collage_report.md .